

PARAMETER ROBUST DESIGNS FOR IMMUNOASSAYS: AN ELISA EXAMPLE

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Summary: A parameter robust design for an immunoassay, such as the ELISA, is one that is not too sensitive to misspecification of the parameter values of the assumed regression model, and predicts a wide range of the unknown concentrations of a specified biochemical substance as precisely as possible. In this paper we use the ELISA example to demonstrate that the parameter robust optimality such as the weighted minimax and the expectation criteria have the potential of obtaining reasonably efficient parameter robust designs for immunoassays.

1. Introduction

Immunoassays such as the radio immunoassays (RIA) and the enzyme-linked immunosorbent assays (ELISA) are techniques for detecting and/or quantifying biochemical substances in certain media, e.g. antibodies in body fluids. Basically, the biochemical analyses of the samples with known (z)

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and unknown concentrations of the substance (z^*) give rise to responses y and y^* , respectively, which are related to their respective concentrations by similar nonlinear regression models. Then regression methods such as the classical calibration method (Osborne (1991)) are used to predict the unknown concentration of the substance (z^*) in the samples.

U	U	S	S	U	U	U	U	U	S	U	S
S	S	U	U	S	S	U	U	U	U	S	S
S	S	S	U	U	S	U	U	U	U	U	U
S	S	U	U	U	U	U	U	U	U	U	U
U	U	S	U	U	U	S	S	U	S	S	S
U	S	S	S	U	S	U	U	U	U	U	U
S	U	S	U	U	U	U	U	U	S	U	U
S	U	S	U	U	U	U	U	U	S	S	U

Figure 1. A 96-well microplate for the RIA or the ELISA. U are wells for the unknowns; S are wells for the standards.

The design of the assays is as follows (Rocke and Jones (1997)). Samples are analysed in wells of a fixed K -well microplate (figure 1) that is divided into wells for the *standards* and wells for the *unknowns*. The standards are k samples ($k \geq \#(\text{model parameters})$) with different but known concentrations (z) of the substance. Each standard is analysed in n_i ($i = 1, 2, \dots, k$) of the K wells to obtain the standard sample $S = \{y_{ij}, z_i\}_{j=1, i=1}^{n_i, k}$ for fitting the regression model. The unknowns are k^* individual samples with unknown concentrations (z^*) of the substance. Each of the k^* unknowns is analysed as is in n^* of the remaining $k^*n^* = K - N$ ($N = \sum_{i=1}^k n_i$) wells if the unknowns are to be predicted using the classical method. This yields the

unknown samples $U = \{y_{i'}^*, z^*\}_{i'=1}^{n^*}$. Thus, the exact design for predicting the unknowns using the classical calibration method is given by the array

$$\xi_K = \left\{ \begin{array}{cccc|cc} z_1 & z_2 & \dots & z_k & S & U \\ n_1 & n_2 & \dots & n_k & N & K - N \end{array} \right\} = \{\xi_K^z | \xi_K^c\} \quad (1)$$

If for K large enough,

$N = (1 - k^*q)K$, $q \in [1/K, 1/k^*]$, $n_i = p_i N$, $p_i \in [0, 1]$, $\sum_{i=1}^k p_i = 1$, then the continuous counterpart (Silvey, 1980, p. 13) of design ξ_K is given by

$$\xi = \left\{ \begin{array}{cccc|cc} z_1 & z_2 & \dots & z_k & S & U \\ p_1 & p_2 & \dots & p_k & 1 - k^*q & k^*q \end{array} \right\} = \{\xi^z | \xi^c\} \quad (2)$$

In design ξ , ξ^c is a design measure which allocates a proportion $(1 - k^*q)$ of the K wells to the standards (S) and equally divides the remainder between the k^* unknowns (U), and ξ^z is a design measure which allocates a proportion p_i of the wells allocated the standards to the standard z_i . This paper is about the construction of design ξ which is insensitive to misspecification of the model parameters and predicts a wide range of the unknowns as precisely as possible.

The assumption of the classical calibration method is that the standard sample (S) and the unknown samples (U) are described by the respective parallel nonlinear regression models

$$y_{ij} = f(\phi, z_i) + f^\theta(\phi, z_i)\epsilon_{ij}, \quad \epsilon_{ij} \text{ iid } N(0, \sigma^2) \quad (3)$$

$$y_{i'}^* = f(\phi, z^*) + f^\theta(\phi, z^*)\epsilon_{i'}, \quad \epsilon_{i'} \text{ iid } N(0, \sigma^2) \quad (4)$$

where the ϵ_{ij} and the $\epsilon_{i'}$ are independent random errors, θ is known, $\phi \in \Phi \subset R^p$ ($p \leq k$) is a parameter vector, and $f(\phi, z^*)$ is an invertible mean response function of $z^* \in \Xi^* \subset R$. Then ϕ is estimated from S using the maximum likelihood method, and the predicted value of z^* by the

classical calibration method is $\hat{z}^*$ which solves the equation (provided the solution exists)

$$\bar{y}^* = \frac{1}{n^*} \sum_{i=1}^{n^*} y_{i'}^* = f(\hat{\phi}, \hat{z}^*) \quad (5)$$

where $\hat{\phi}$ is the maximum likelihood estimate of ϕ . If $\hat{z}^*$ is a differentiable function of $\hat{\phi}$ and $\bar{y}^*$, then the standardized asymptotic variance of $\hat{z}^*$ is given by

$$\begin{aligned} V(\phi, z^*, \xi) &= \left[\frac{\partial \hat{z}^*}{\partial \hat{\phi}'} M^{-1}(\phi, \xi^z) \frac{\partial \hat{z}^*}{\partial \hat{\phi}} \frac{f^{2\theta}(\phi, z^*)}{q} \left(\frac{\partial \hat{z}^*}{\partial \bar{y}^*} \right)^2 \right]_{\hat{\phi}=\phi, \bar{y}^*=f(\phi, z^*)} \\ &= \frac{v_1(\phi, z^*, \xi^z)}{1 - k^*q} + \frac{k^*v_2(\phi, z^*)}{k^*q} \end{aligned} \quad (6)$$

where $M(\phi, \xi^z)$ is the average Fisher information matrix for ϕ given by

$$M(\phi, \xi^z) = \sum_{i=1}^k p_i \left(\frac{1}{f^{2\theta}(\phi, z_i)} + \frac{2\theta^2}{f^2(\phi, z_i)} \right) \frac{\partial f(\phi, z_i)}{\partial \phi} \frac{\partial f(\phi, z_i)}{\partial \phi'} \quad (7)$$

(Downing, Fedorov and Leonov, 2001, p. 45–51).

In (6) $\frac{1}{1-k^*q}v_1(\phi, z^*, \xi^z)$ and $\frac{1}{q}v_2(\phi, z^*)$ are the components of variance due to the random errors in the standard and the unknown samples, respectively. The following ELISA example shows that neither component of variance (6) (or of design (2)) can be ignored when designing immunoassays. Furthermore, the example shows that (6) can be an unbounded function of $(\phi', z^*)'$. Hence, designs ξ which minimise (6) for all $(\phi', z^*)' \in \Phi \times \Xi^*$ may not exist or may be locally optimal with respect to a few values of $(\phi', z^*)'$ associated with large $V(\phi, z^*, \xi)$.

ELISA Example: For the ELISA (Rocke and Jones, 1997, Downing et al., 2001, p. 45–51), y is the optical density, z is the *log* concentration of the

substance, $\theta = 1$,

$$f(\phi, z) = \phi_4 + \frac{\phi_1}{1 + e^{-\phi_2(z - \phi_3)}}, \phi_1 > 0, \phi_2 \neq 0, |\phi_3| < \infty, \phi_4 \geq 0 \quad (8)$$

$$\hat{z}^* = \hat{\phi}_3 + \frac{1}{\hat{\phi}_2} \ln \left\{ \frac{\bar{y}^* - \hat{\phi}_4}{\hat{\phi}_1 + \hat{\phi}_4 - \bar{y}^*} \right\}, \bar{y}^* \in (\hat{\phi}_4, \hat{\phi}_1 + \hat{\phi}_4), |\hat{\phi}_2| \neq 0 \quad (9)$$

$$M(\phi, \xi^z) = \sum_{i=1}^k \frac{3p_i}{f^2(\phi, z_i)} \frac{\partial f(\phi, z_i)}{\partial \phi} \frac{\partial f(\phi, z_i)}{\partial \phi'}. \quad (10)$$

Furthermore, for $(k^*, \phi_1, \phi_2, \phi_3, \phi_4) = (20, .35, -.872, 4.662, .151)$ one parameter robust design for the ELISA which we calculated using formulae provided by Rocke and Jones (1997) is

$$\xi^r = \left\{ \begin{array}{ccccc|cc} 2.2259 & 3.0999 & 3.9739 & 4.8479 & 5.7219 & S & U \\ .2 & .2 & .2 & .2 & .2 & .2 & .8 \end{array} \right\} = \{\xi^z | \xi^c\}. \quad (11)$$

With this design 20% of the wells of a K -well microplate are equally divided among the 5 standards, and the remaining wells are equally divided among the 20 unknowns. The graph of $V(\phi, z^*, \xi^r)$ versus z^* (figure 2) shows that both components of variance (6) contribute significantly to $V(\phi, z^*, \xi)$, and are unbounded functions of $z^* \in R$ due to the sigmoidal shape of the mean response function (8) on R .

One approach to constructing designs for the ELISA type of immunoassays is discussed in the paper by Rocke and Jones (1997) which improves on a paper by Bunch, Rocke and Harrison (1990). In this approach the user chooses a convenient number (k) of equally replicated standards and the desired number (k^*) of equally replicated unknowns to be analysed in a microplate. Then the standards are serial dilutions of a standard sample (midpoint of the dilution series) which with the dilution ratio are chosen to maximise $\int_R 1/V(\phi, z^*, \xi) dz^*$ evaluated at the user specified model parameter values.

The authors report this approach obtains designs for the ELISA and the RIA that are not too sensitive to misspecification of the model parameter values.

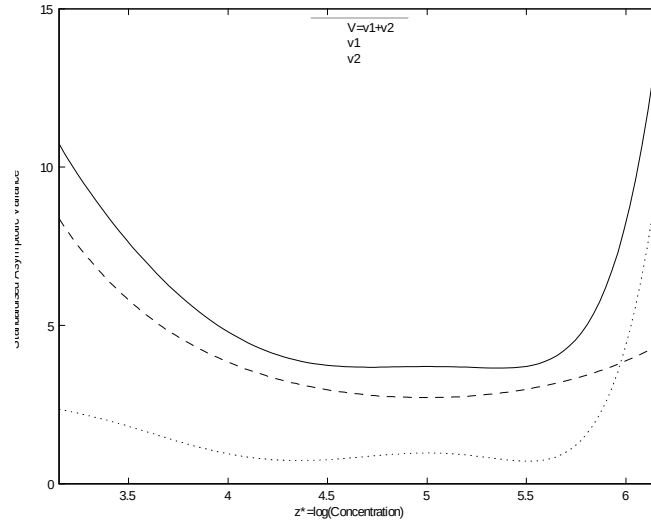


Figure 2. The components of the standardised asymptotic variance of $\hat{z}^*$ – the ELISA case: $V = V(\phi, z^*, \xi)$; $v1 = \frac{1}{1-k^*q}v_1(\phi, z^*, \xi^z)$; $v2 = \frac{1}{q}v_2(\phi, z^*)$; $(k^*, \sigma, \phi_1, \phi_2, \phi_3, \phi_4) = (20, .045, .35, -.872, 4.662, .151)$; and design (11) of Rocke and Jones (1997).

A competing alternative design approach is to use parameter robust optimality criteria such as the minimax and the expectation / Bayesian criteria (Ford, Titterton and Kitsos (1989), Chaloner and Larntz (1989)) to construct designs ξ . One possible advantage of this approach over that of Rocke and Jones (1997) is the possible reduction of the number of user specified design parameters which may result in designs which are more efficient than those of the authors. However, the equivalence theorems associated with the criteria are not directly applicable in this case (because ξ is composed of two components ξ^z and ξ^c with different supports) and hence need to be extended

to calculating and/or checking the optimality of the designs ξ . We are not aware of the application of the criteria to constructing designs ξ although the criteria have been used to constructing the ξ^z component of ξ for precisely estimating functions of the model parameters, e.g. Chaloner and Larntz (1989), Haines (1995), and Dette and Sahm (1998).

The objective of this paper is to demonstrate that the existing parameter robust optimality criteria can obtain parameter robust designs ξ that are reasonably efficient for predicting a wide range of the unknowns. In particular we demonstrate that the weighted minimax and the expectation criteria can obtain reasonably efficient designs (relative to the designs of Rocke and Jones (1997)) for the ELISA. The paper is organised as follows. In section 2.1 we list the assumptions and the specifications of the design for the ELISA. In sections 2.2 and 2.3 we discuss the construction of the expectation and the weighted minimax designs, respectively. In each case we define and construct the designs, illustrate how to check the optimality of the designs, and evaluate the efficiency of the designs relative to the design of Rocke and Jones (1997). Section 3 is the conclusion.

2. Expectation and Minimax Designs for the ELISA

2.1 Assumptions and Specifications

In the sections that follow we explore the application of the expectation and the weighted minimax criteria to constructing designs for the ELISA using models (3) and (4) with (8) – (10), and the prior parameter space given by

$$\begin{aligned} \Phi &= \{.2765, .3500, .4235\} \times \{-1.1194, -.8720, -.6246\} \\ &\times \{4.1547, 4.6620, 5.1693\} \times \{.1086, .1510, .1934\}. \end{aligned} \quad (12)$$

The middle parameter values $\phi_1 = .35$, $\phi_2 = -.872$, $\phi_3 = 4.662$ and $\phi_4 = .151$ are equivalent to those used by Rocke and Jones (1997). The boundary parameter values are supposed to be the user's best guess at a parameter value $\pm$ three standard deviations (which should be possible to guess for routine immunoassays).

The classical calibration method is most efficient for predicting the unknown z^* that are within region Ξ^* on which (8) is approximately a linear function of z^*/z . This region is given by

$$\Xi^* = [\phi_3 + \frac{1}{|\phi_2|} \ln\{\frac{3 - \sqrt{3}}{3 + \sqrt{3}}\}; \phi_3 + \frac{1}{|\phi_2|} \ln\{\frac{3 + \sqrt{3}}{3 - \sqrt{3}}\}] \quad (13)$$

where the limits of Ξ^* are solutions for z in the equation $\frac{\partial^3 f(\phi, z)}{\partial z^3} = 0$. Hence, for each (ϕ_2, ϕ_3) we take Ξ^* to be the region of the predictable unknown z^* . The actual support for the ξ^z component of design ξ is R but we take the widest Ξ^* , $\Xi = [2.0462, 7.2778]$, to be the support for ξ^z .

There is need to specify the upper limit for q , $1/k^*$ in design ξ , in order to avoid numerical instabilities during the optimisation process. Equation (6) shows that $V(\phi, z^*, \xi)$ increases with number of unknowns k^* . Hence the choice of k^* will always depend on the user's preference for precision versus the number of unknowns per microplate (cost according to Rocke and Jones, 1997). We illustrate the construction of the designs with $k^* = 1$, 20.

2.2 Expectation Designs

Let $P(\phi, z^*)$ be a probability distribution defined on $\Phi \times \Xi^*$ such that

$$AV(\xi) = E_P[V(\phi, z^*, \xi)] = \frac{E_P[v_1(\phi, z^*, \xi^z)]}{1 - k^*q} + \frac{E_P[v_2(\phi, z^*)]}{q} < \infty. \quad (14)$$

Then design ξ^1 is an optimal expectation design with respect to the probability distribution $P(\phi, z^*)$ if

$$AV(\xi^1) = \min_{\xi} AV(\xi). \quad (15)$$

It is clear from (14) that ξ^1 can be constructed by first optimising ξ^z and then calculating the optimal q using the formula

$$q_1 = \frac{\sqrt{E_P[v_2(\phi, z^*)]}}{\sqrt{k^*E_P[v_1(\phi, z^*, \xi^z)]} + k^*\sqrt{E_P[v_2(\phi, z^*)]}} \quad (16)$$

for all positive integers k^* . Hence, design ξ^1 is optimal if the directional derivative

$$\begin{aligned} D_1(z) &= E_P[v_1(\phi, z^*, \xi^z)] - \\ &\quad E_P \left[\frac{\partial \hat{z}^*}{\partial \hat{\phi}} M^{-1}(\phi, \xi^z) M(\phi, z) M^{-1}(\phi, \xi^z) \frac{\partial \hat{z}^*}{\partial \hat{\phi}} \bigg|_{\hat{\phi}=\phi, \bar{y}^*=f(\phi, z^*)} \right] \\ &\geq 0 \end{aligned} \quad (17)$$

for all $z \in \Xi$ with the equality holding if z is the support point of ξ^z , and ξ^c is given by (16). This is an extension of the equivalence theorems in Chaloner and Larntz (1989), and in Whittle (1973).

A suitable choice of $P(\phi, z^*)$ is one that puts more mass on what are perceived to be the most likely values of $(\phi', z^*)'$. For example, a local optimal design ξ^1 can be obtained by choosing $P(\phi, z^*)$ degenerate at a specified value of $(\phi', z^*)'$. In the absence of the knowledge about the likely values of $(\phi', z^*)'$, a reasonable choice of $P(\phi, z^*)$ is given by a uniform distribution for $\phi \in \Phi$ and a conditional uniform distribution for $z^* \in \Xi^*$.

These uniform distributions reflect ignorance about the locations of ϕ and z^* in Φ and Ξ^* , respectively.

We used the “fmincon” optimization routine in Matlab Version 5.3 (1996) to construct the expectation design for $P(\phi, z^*)$ defined by the above two uniform distributions. The optimal designs were

$$\xi^1 = \left\{ \begin{array}{cccc|cc} 2.0462 & 3.8880 & 5.732 & 7.2778 & S & U \\ .1659 & .4295 & .3146 & .0900 & 1 - k^*q & k^*q \end{array} \right\} \quad (18)$$

with the optimal q given by (16) as

$$q_1 = \frac{1}{\sqrt{1.1384k^*} + k^*} = \begin{cases} .4838 & \text{if } k^* = 1 \\ .0404 & \text{if } k^* = 20 \end{cases} \quad (19)$$

This means if $k^* = 1$, then 48.38% of the wells of a K -well microplate are allocated to the single unknown, and the remaining wells are allocated to the 4 standards according to the weights of the ξ^z component of design ξ^1 . Otherwise if $k^* = 20$, then 80.80% of the wells of the microplate are equally divided between the 20 unknowns, and the remaining wells are allocated to the 4 standards according to the weights of ξ^z . Figure 3 confirms the optimality of the expectation designs given by (18) and (19). Figure 4 shows that, on average, design (18) for $k^* = 20$ is more efficient than design (11) of Rocke and Jones (1997).

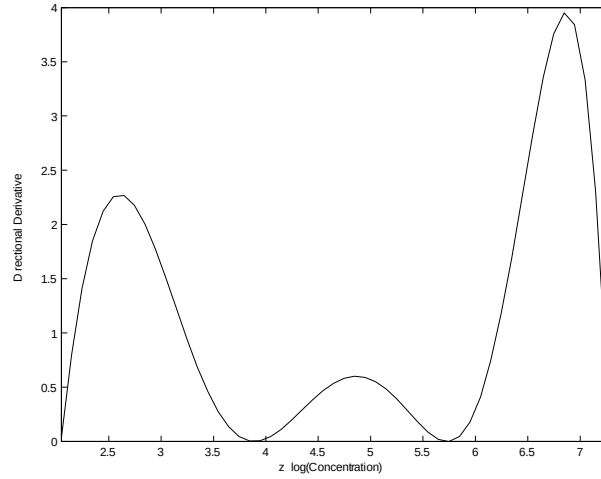


Figure 3. The directional derivative function $D_1(z)$ (17) associated with the expectation designs (18).

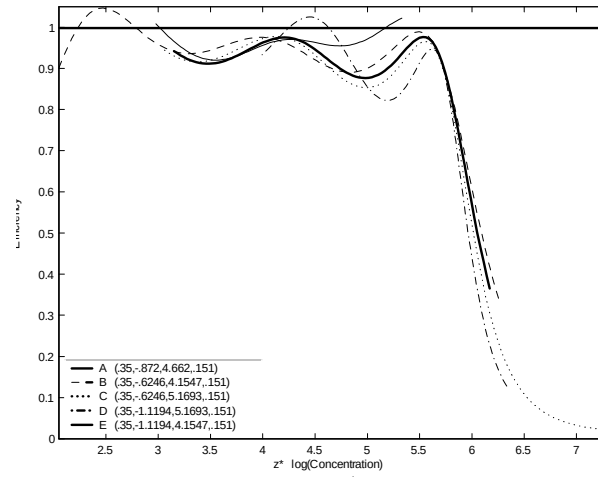


Figure 4. The efficiency $(= V(\phi, z^*, \xi^1)/V(\phi, z^*, \xi^r))$ of the expectation design (18) for $k^* = 20$ relative to design (11) of Rocke and Jones (1997). $(\phi_1, \phi_2, \phi_3, \phi_4) = A, B, C, D$.

The overall conclusion of this section is that the expectation criterion can define reasonably efficient parameter robust designs for nonlinear classical calibration. The expectation criterion has the nice property that the optimal ξ^z component of the designs does not depend on the ξ^c component. Furthermore, for each optimal ξ^z there is a simple formula (16) for calculating the optimal ξ^c component of the designs. The choice of k^* and the distribution $P(\phi, z^*)$ determine the range of both the efficiency and the robustness of the designs. While the choice of $P(\phi, z^*)$ has the Bayesian support, the choice of k^* is up to the user's preference for precision versus the number of unknowns to be analysed in a microplate (as was mentioned in section 2.1). For example, a user may choose k^* which satisfies either a desired precision constraint, or a desired cost constraint, or both.

2.3 Weighted Minimax Designs

Let $W(\phi, z^*)$ be a nonnegative weight function defined on $\Phi \times \Xi^*$ such that

$$\varphi(\phi, z^*, \xi) = W(\phi, z^*)V(\phi, z^*, \xi) < \infty. \quad (20)$$

Then design ξ^2 is an optimal weighted minimax design with respect to the weight function $W(\phi, z^*)$ if

$$\varphi(\phi, z^*, \xi^2) = \min_{\xi} \max_{(\phi', z^*)' \in \Phi \times \Xi^*} \varphi(\phi, z^*, \xi). \quad (21)$$

Equivalently, let $\{(\phi'_i, z^*_i)'\}$ be the set of values of $(\phi', z^*)'$ for which

$$\varphi(\phi, z^*, \xi) = \max_{(\phi', z^*)' \in \Phi \times \Xi^*} \varphi(\phi, z^*, \xi).$$

Then design ξ^2 is optimal if there exists nonnegative multipliers ν_i with sum equal to 1 such that the “directional derivative”

$$D_2(z) = \sum_i \nu_i \frac{\frac{\partial \hat{z}^*}{\partial \phi} M^{-1}(\phi_i, \xi^z) M(\phi_i, z) M^{-1}(\phi_i, \xi^z) \frac{\partial \hat{z}^*}{\partial \phi}}{\frac{\partial \hat{z}^*}{\partial \phi} M^{-1}(\phi_i, \xi^z) \frac{\partial \hat{z}^*}{\partial \phi}} \bigg|_{\hat{\phi}=\phi_i, \bar{y}^*=f(\phi_i, z_i^*)} \leq 1 \quad (22)$$

for all $z \in \Xi$ with the equality holding if z is the support point of ξ^z , and the optimal q is given by

$$q_2 = \frac{\sum_i \nu_i W(\phi_i, z_i^*) v_2(\phi_i, z_i^*)}{\sqrt{k^* \sum_i \nu_i W(\phi_i, z_i^*) v_1(\phi_i, z_i^*, \xi^z) + k^* \sqrt{\sum_i \nu_i W(\phi_i, z_i^*) v_2(\phi_i, z_i^*)}}} \quad (23)$$

for each positive integer k^* . This is an extension of the equivalence theorems in Dette and Sahm (1998), and Pukelsheim (1993), p. 175. In this case an inspection of $\varphi(\phi, z^*, \xi)$ shows that the optimal ξ^z and the optimal ξ^c are interdependent unless $W(\phi, z^*) v_2(\phi, z^*)$ does not depend on $(\phi', z^*)'$ (in which case the optimal ξ^z does not depend on ξ^c as with expectation criterion).

The purpose of $W(\phi, z^*)$ is to moderate the variability of $V(\phi, z^*, \xi)$ on $\Phi \times \Xi^*$ so that the designs defined by the weighted minimax criterion are not locally optimal with respect to a few values of $(\phi', z^*)'$ associated with large $V(\phi, z^*, \xi)$. The choice of such $W(\phi, z^*)$ on $\Phi \times \Xi^*$ is subjective, e.g. Bunch et al. (1990) use a Gaussian weight function defined on Ξ^* . However, the usual minimax criterion is obtained if $W(\phi, z^*) = 1$, and the efficiency criterion (Ford et al., 1989) is obtained if $W(\phi, z^*) = 1/V(\phi, z^*, \xi^r)$ where ξ^r is some reference design.

For illustration purposes we used the “fminimax” optimization routine in Matlab Version 5.3 (1996) to construct the weighted minimax designs defined by the weight functions $W(\phi, z^*) = 1$ and $W(\phi, z^*) = 1/V(\phi, z^*, \xi^r)$. Table 1 displays the designs. Here the optimal ξ^z component of the designs

depends on k^* because $W(\phi, z^*)v_2(\phi, z^*)$ depends on $(\phi', z^*)'$. Figure 5 shows that the designs are near optimal. Figures 6 and 7 show the effect of the choice of the weight function on the efficiency of the designs. The design for $W(\phi, z^*) = 1$ is less efficient than the design of Rocke and Jones (1997) because the ξ^z component of the design is too right skewed due to the fact that $V(\phi, z^*, \xi)$ is larger at low concentrations than at high concentrations of the unknown. On the other hand, the design for $W(\phi, z^*) = 1/V(\phi, z^*, \xi^r)$ is on average more efficient than the design of Rocke and Jones (1997) because the weight function moderates the variability of $V(\phi, z^*, \xi)$ on $\Phi \times \Xi^*$.

This section has demonstrated that the weighted minimax criterion can also define reasonably efficient parameter robust designs for nonlinear classical calibration although the application of this criterion is not as straightforward as that of the expectation criterion. Firstly, the choice of $W(\phi, z^*)$ requires the user to have some knowledge of the behaviour of $V(\phi, z^*, \xi)$ on $\Phi \times \Xi^*$ so that he/she can choose a weight function for which the weighted variance has comparable values on $\Phi \times \Xi^*$. Otherwise users should try $1/V(\phi, z^*, \xi^r)$ or $1/v_2(\phi, z^*)$ as weight functions. (The designs for the latter weight function also have the property that the optimal ξ^z component of the designs does not depend on the ξ component as was the case with the expectation designs).

Table 1. The weighted minimax designs for the weight functions $W(\phi, z^*) = 1$ and $1/V(\phi, z^*, \boldsymbol{\xi}^r)$ on $\Phi \times \Xi^*$ (k^* = number of unknowns). $\boldsymbol{\xi}^r$ is the design (11) of Rocke and Jones (1997).

$W(\phi, z^*)$		Design $\boldsymbol{\xi}^2$									
		k^*	z_1	z_2	z_3	z_4	z_5	S	$1 - k^*q$	S	k^*q
1	1	1	2.0462	3.6011	5.9754	7.2778					
			.4405	.3022	.1157	.1416					
		20	2.0462	3.5281	6.0483	7.2778					
			.5002	.3567	.0653	.0778					
$1/V(\phi, z^*, \boldsymbol{\xi}^r)$	20	20	2.0462	3.3721	4.5677	5.7009	7.2778				
			.2107	.2713	.2645	2.354	.0181				

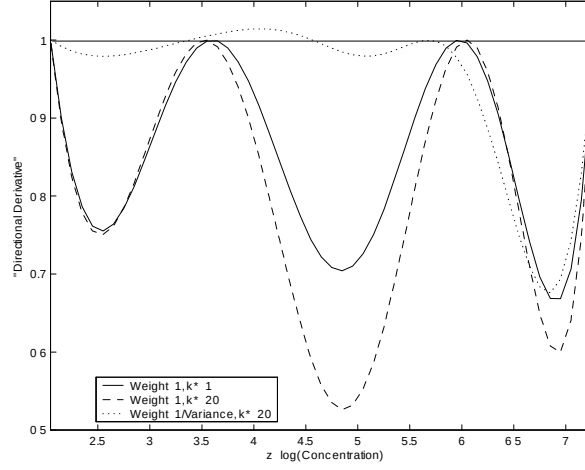


Figure 5. The “directional derivative” functions associated with the weighted minimax designs in table 1. $\text{Weight} = 1 \equiv W(\phi, z^*) = 1$; $\text{Weight} = 1/\text{Variance} \equiv W(\phi, z^*) = 1/V(\phi, z^*, \xi^r)$; ξ^r is design (11) of Rocke and Jones (1997).

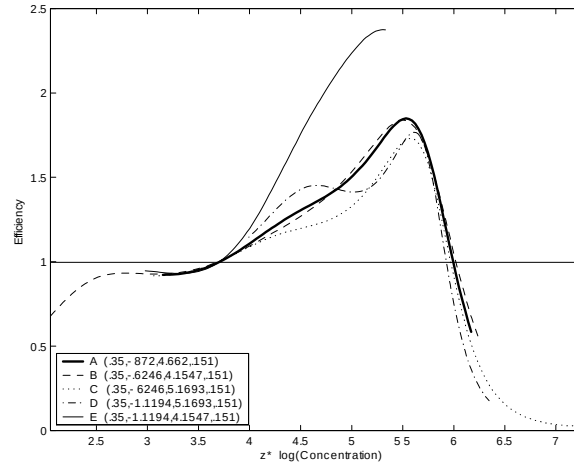


Figure 6. The efficiency $(= V(\phi, z^*, \xi^2)/V(\phi, z^*, \xi^r))$ of the weighted minimax design for the weight function $W(\phi, z^*) = 1$ and $k^* = 20$ (table 1) relative to design (11) of Rocke and Jones (1997). $(\phi_1, \phi_2, \phi_3, \phi_4) = A, B, C, D$.

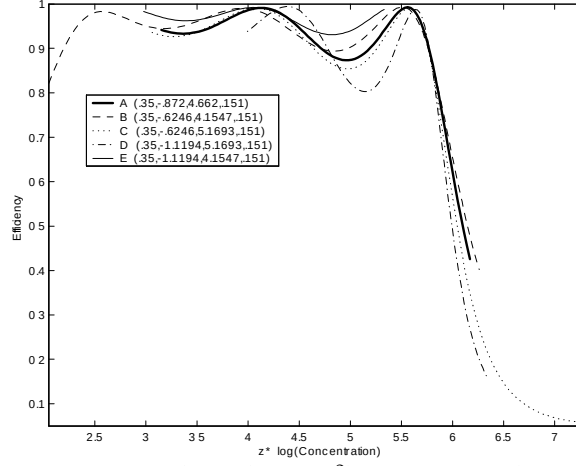


Figure 7. The efficiency $(= V(\phi, z^*, \xi^2) / V(\phi, z^*, \xi^r))$ of the weighted minimax design for the weight function $W(\phi, z^*) = 1/V(\phi, z^*, \xi^r)$ and $k^* = 20$ (table 1) relative to design (11) of Rocke and Jones (1997). $(\phi_1, \phi_2, \phi_3, \phi_4) = A, B, C, D$.

Users may also choose $W(\phi, z^*)$ based on prior information about the most likely values of $(\phi', z^*)' \in \Phi \times \Xi^*$. Secondly, checking the optimality of the weighted minimax designs is more complicated as it involves checking the existence of the nonnegative multipliers ν_i , in (22), which solve systems of nonlinear equations. As with the expectation designs, the choice of k^* and the weight function $W(\phi, z^*)$ determine the range of both the efficiency and the robustness of the designs. The choice of k^* is up to the user's preference for precision versus the desired number of unknowns to be analysed in a microplate as was mentioned in sections 2.1 and 2.2.

3. Conclusion

The numerical results in this paper have shown that the weighted minimax and the expectation criteria can define parameter robust designs which are

reasonably efficient for nonlinear calibration. In particular, the results show that the weighted minimax and the expectation criteria have the potential of obtaining designs for the ELISA which are more efficient than those of Rocke and Jones (1997). However, the designs of Rocke and Jones (1997) are easier to calculate (the authors provide simple formulae for calculating the designs by hand) than the weighted minimax and the expectation designs whose construction requires specialised numerical routines, e.g. for integration and optimisation. The construction of the minimax and the expectation designs is further complicated by the need for checking the optimality of the designs, which is not automatically done by most optimisation routines. We found checking the optimality of the designs necessary as the Matlab optimisation routines very often converged at non-optimal designs.

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